

Physics-Informed Joint-Embedding Predictive Architectures for Multivariate Time-Series Anomaly Detection: Spectral, Transport, and Field-Theoretic Formulations

Andre Williams	Imadeldin Mahgoub	Hamid Akbarian
<i>Florida Atlantic University</i>	<i>Florida Atlantic University</i>	<i>Florida Atlantic University</i>
<i>Department of EECS</i>	<i>Department of EECS</i>	<i>Department of EECS</i>
<i>Boca Raton, US</i>	<i>Boca Raton, US</i>	<i>Boca Raton, US</i>
<i>andrewwilliam2018@fau.edu</i>	<i>mahgoubi@fau.edu</i>	<i>hakbaria@fau.edu</i>

Abstract—Unsupervised anomaly detection in multivariate telemetry is central to operational safety across aerospace systems, industrial processes, cyber-physical infrastructure, and smart environments. While reconstructive architectures such as TimesNet and TranAD achieve competitive continuous ranking performance (PR-AUC 0.2585–0.2612), their observation-space residuals destabilize automated streaming threshold calibration, generating over 1.04×10^6 false positives. To address this calibration failure, we evaluate Joint-Embedding Predictive Architectures (JEPAs) that forecast dynamics in regularized latent representation space. We introduce three physics-inspired inductive training regularizers grounded in dynamical systems theory, continuum mechanics, and field theory: (1) *Operator-Entropy JEPA*, which parameterizes a latent transition matrix regularized by Von Neumann spectral entropy during pretraining to prevent modal collapse; (2) *Reynolds-Stress JEPA*, which adapts hydrodynamic Reynolds decomposition to align second-order fluctuation covariance across latent coordinates; and (3) *Potential-Flow JEPA*, which parameterizes latent trajectory transitions via an irrotational velocity potential ($\hat{v} = \nabla\Phi$). To handle heterogeneous sensor arrays, the framework incorporates coordinate saliency gating for active channel isolation, dyadic wavelet shells for multi-scale frequency decomposition, and kurtosis-adapted Extreme Value Theory (EVT) for streaming threshold calibration. Evaluated on a matched benchmark of 45 continuous telemetry channels across 11 datasets, latent prediction suppresses false alarms while capturing true physical anomalies. Operator-Entropy JEPA achieves the highest Dataset-Macro Point-F1 (0.2066, a +43.2% relative improvement over TimesNet and +28.4% over TranAD) and is the only evaluated architecture to exceed the analytical All-Positive baseline on Dataset-Macro Point-F1 (0.1893), though on Channel-Macro all deep models fall below the constant alarm baseline (0.1942) due to extreme channel-level sparsity. In Pooled Micro Point-F1, Potential-Flow JEPA achieves 0.3458 ($3.2\times$ higher than TranAD and $5.3\times$ higher than TimesNet) while reducing false positives by 8.0-fold against TimesNet (130,237 vs 1,047,144) and 3.5-fold against TranAD (452,621). These findings establish that latent predictive architectures yield better-conditioned residual distributions that substantially mitigate streaming threshold calibration collapse, with physical inductive regularizers providing targeted performance gains in specialized dynamical regimes.

Index Terms—Multivariate Time-Series, Anomaly Detection, Joint-Embedding Predictive Architecture (JEPA), Physics-Inspired Machine Learning, Spectral Entropy, Reynolds Stress, Potential Flow, Extreme Value Theory, Deep Learning.

I. INTRODUCTION

Autonomous anomaly detection in multi-sensor telemetry monitors operational health across aerospace systems, Earth observation, cyber-physical infrastructure, and wearable digital healthcare [1], [2]. In remote sensing deployments (such as the NASA Soil Moisture Active Passive (SMAP) satellite, Mars Science Laboratory (MSL) rovers, industrial IoT facilities, and wearable biometric sensor arrays), dozens to hundreds of continuous telemetry channels concurrently track electrical power buses, thermal gradients, mechanical vibrations, and kinematics [3]. When sensor faults or operational breakdowns emerge at the edge, monitoring systems must identify dynamical anomalies before hardware damage occurs [4].

A. Challenges in Modeling Multi-Sensor Telemetry

From a physical perspective, multi-sensor telemetry reflects the state trajectory of an underlying continuous physical system $\dot{x}(t) = f(x(t))$. However, attempting to model dynamical equations directly in raw observation space presents four fundamental challenges on edge devices. First, estimating instantaneous time derivatives $\dot{x} \approx \frac{\Delta x}{\Delta t}$ directly on discrete, noise-corrupted sensor streams amplifies high-frequency measurement noise, destabilizing numerical trajectory tracking. Second, remote sensing telemetry captures only a noisy surface projection of unmeasured thermodynamic, electrical, and environmental states, violating full-state observability. Third, in large sensor arrays ($K \geq 50$), cross-channel dependencies and non-linear interactions cannot be adequately represented by rigid functional forms in raw signal space. Finally, edge IoT and satellite systems undergo discrete operational mode transitions (e.g., orbital day/night thermal cycling, thruster firings, Parkinsonian gait freezing, cloud server garbage collection) that cause continuous distribution drift.

B. Limitations of Deep Anomaly Detectors

Deep learning architectures provide data-driven alternatives to explicit equation modeling, yet they face three primary

limitations. First, generative models (e.g., autoencoders, VAEs, masked Transformers) minimize point-level L_1/L_2 errors in raw signal space [3], [5], [6]. Because reconstruction loss penalizes every discrepancy equally, autoencoders allocate capacity to fitting high-frequency sensor jitter rather than learning nominal dynamical transitions. Second, contrastive frameworks such as Neural Contextual Anomaly Detection (NCAD) [7] train encoders via Contextual Outlier Exposure (COE) using heuristic synthetic perturbations (e.g., Gaussian noise spikes, sequence splicing). However, real physical faults manifest as subtle deviations from nominal dynamics; synthetic injections distort embedding geometry, inducing artificial decision boundaries that generate false alarms on continuous telemetry. Third, in high-dimensional observation space, even after per-channel standardization, Euclidean reconstruction errors ($\|x - \hat{x}\|_2^2 = \sum_k (x_k - \hat{x}_k)^2$) accumulate stochastic residual variance across dozens of irrelevant or noisy sensors. Under automated streaming thresholding such as Peaks-Over-Threshold (SPOT), this residual accumulation yields heavy, unstable tail distributions that distort extreme-value calibrators, triggering false alarm flooding in operational telemetry.

C. Our Approach: Physics-Inspired Joint-Embedding Predictive Architectures

To substantially suppress sensitivity to observation-space noise without relying on synthetic contrastive perturbations, we formulate a family of *Physics-Inspired Joint-Embedding Predictive Architectures* (PI-JEPA) for multi-sensor telemetry. Instead of reconstructing raw, jitter-corrupted sensor observations $\hat{x} \approx x$, a JEPA maps past observations x_{ctx} into an abstract representation space and predicts target representations z_{tgt} directly [8], [9]. By operating in latent representation space, the architecture filters out high-frequency stochastic sensor noise that lacks long-range dynamical predictability.

However, unconstrained latent prediction alone can be underspecified in complex multi-sensor telemetry. To incorporate physical domain priors into self-supervised representation learning, we introduce three physics-inspired JEPA architectures featuring inductive dynamical regularizers. The first formulation, *Operator-Entropy JEPA*, parameterizes a latent transition matrix regularized by Von Neumann spectral entropy during pretraining to penalize dimensional collapse and preserve linear modal expressivity in multi-sensor dynamics. The second, *Reynolds-Stress JEPA*, adapts hydrodynamic Reynolds decomposition ($z = \bar{z} + z'$) to align second-order fluctuation covariance across latent coordinates, capturing structural anomalies that disrupt multi-coordinate covariance balance. The third, *Potential-Flow JEPA*, enforces conservative latent trajectory transitions by parameterizing the transition field as the gradient of a scalar potential ($\hat{v} = \nabla\Phi$); because gradient fields are irrotational by construction, non-conservative dynamical anomalies produce pronounced predictive discrepancies.

To stabilize learning across heterogeneous high-dimensional sensor arrays, all three architectures incorporate Coordinate Saliency Gating to isolate active channels, Dyadic Wavelet Shells to decouple high-frequency transients from baseline drift, and a Kurtosis-Adaptive Extreme Value Theory (EVT

/ SPOT) calibrator that modulates tail thresholds against non-Gaussian kurtosis.

D. Contributions

The primary contributions of this paper are organized as follows:

1) Physics-Inspired Joint-Embedding Framework: We formulate an unsupervised anomaly detection framework that embeds physical inductive regularizers (spectral dissipation, turbulent momentum transfer, conservative flow) directly into latent representations, avoiding raw-space noise reconstruction.

2) Flagship Inductive Formulations: We derive and implement three physics-inspired architectures (Operator-Entropy JEPA, Reynolds-Stress JEPA, and Potential-Flow JEPA) and benchmark them systematically alongside an unconstrained control (TS-JEPA).

3) Inductive Architectural Components: We introduce Coordinate Saliency Gating for cross-channel fault localization, Dyadic Wavelet decomposition for multi-resolution temporal analysis, and Kurtosis-Adaptive EVT tail calibration to provide automated streaming alarms without transductive test tuning.

4) Multi-Tier Benchmark Evaluation: We evaluate performance across 11 telemetry datasets (45 matched channels), reporting Dataset-Macro, Channel-Macro, Pooled Micro-F1, and Area Under the Precision-Recall Curve (PR-AUC) alongside an analytical All-Positive reference baseline. We demonstrate that latent prediction mitigates threshold calibration collapse, achieving up to 43.2% relative improvement in Dataset-Macro F1 (where Operator-Entropy JEPA is the sole model to exceed the analytical All-Positive baseline) and a $3.2\times$ to $5.3\times$ gain in Pooled Micro-F1 while reducing false positives by 8.0-fold against TimesNet (130,237 vs 1,047,144) and 3.5-fold against TranAD (452,621).

The remainder of this paper is structured as follows. Section II reviews related work in reconstructive, contrastive, and joint-embedding anomaly detection. Section III details the 11-dataset multi-sensor telemetry benchmark suite. Section IV formalizes the PI-JEPA Triad, coordinate saliency gating, dyadic wavelets, and kurtosis-adaptive EVT. Section V establishes the experimental protocol and multi-tier evaluation framework. Section VI evaluates architectural ablations and inductive layer contributions. Section VII provides domain-by-domain empirical analysis and qualitative physical case studies. Finally, Section VIII concludes the paper.

II. RELATED WORK

Unsupervised time-series anomaly detection spans dynamical equation discovery, reconstructive generative modeling, contrastive representation learning, and predictive world models. This section reviews these approaches and positions our Physics-Informed JEPA framework.

A. Dynamical Systems Modeling and Koopman Operators

The physical modeling of multi-sensor telemetry has long sought to capture system behavior as continuous dynamical

trajectories governed by differential equations $\dot{x}(t) = f(x(t))$ [10]. One prominent approach, Neural Ordinary Differential Equations (Neural ODEs) [11], parameterizes continuous velocity fields using neural networks $\frac{dx}{dt} = f_\theta(x(t), t)$ and integrates trajectories numerically. While Neural ODEs learn flexible vector fields, fitting them directly in raw observation space forces numerical solvers to integrate through high-frequency stochastic sensor noise, leading to numerical stiffness, slow training, and sensitivity to measurement jitter.

In parallel, Koopman operator theory and Dynamic Mode Decomposition (DMD) [12]–[14] analyze non-linear dynamical systems by learning linear transition operators that act on observable functions. Modern deep Koopman frameworks (e.g., Koopa [14]) learn observable embeddings for non-stationary forecasting. In anomaly detection, physical disruptions manifest as spectral decoherence or sudden changes in modal dissipation. Rather than operating on uncalibrated raw signals, our framework parameterizes latent linear transitions regularized by Koopman spectral entropy during training directly in representation space.

B. Reconstruction-Based Detection

The predominant deep learning approach in time-series anomaly detection relies on learning nominal patterns through raw signal reconstruction. Hundman et al. [3] introduced LSTM-based prediction coupled with non-parametric dynamic thresholding for NASA spacecraft telemetry (SMAP/MSL). Modern architectures incorporate specialized attention and frequency-domain transformations. For example, Anomaly Transformer [5] introduces Anomaly-Attention to quantify the association discrepancy between local prior Gaussian kernel associations and global series self-attention. TimesNet [6] decomposes 1D multivariate time series into 2D temporal variations using Fast Fourier Transform (FFT) periodicity discovery and 2D Inception blocks. Similarly, TranAD [15] employs a two-phase adversarial Transformer with sub-series attention and focus scoring to amplify subtle reconstruction errors.

Despite their empirical popularity, reconstructive methods share a fundamental vulnerability: signal-level noise memorization. Because loss functions optimize point-level L_1 or L_2 differences in raw signal space, reconstructive networks are biased toward fitting high-frequency stochastic noise rather than learning invariant dynamical laws. In addition, during sustained anomalies, corrupted observations contaminate the context window, causing the reconstructor to adapt and reconstruct the anomaly as the nominal state.

C. Contrastive Learning and Representation-Based Detection

To avoid reconstructing raw noise, contrastive methods learn discriminative representations by separating nominal sequences from anomalous perturbations. Neural Contextual Anomaly Detection (NCAD) [7] uses Temporal Convolutional Networks (TCNs) trained with Contextual Outlier Exposure (COE), injecting artificial synthetic anomalies (Gaussian noise, point spikes, trend shifts) into training windows to create negative pairs. However, synthetic injections distort the physical

dynamical manifold and fail to generalize to subtle real-world physical faults (such as gradual thermal dissipation or Parkinsonian gait freezing). Alternatively, DCdetector [16] replaces reconstruction with multi-scale dual-attention contrastive representation learning, maximizing cross-scale representation consistency. While DCdetector eliminates artificial injection heuristics, contrastive objectives require careful scale selection and representation pooling.

D. Non-Contrastive Learning and Time-Series JEPAs

Joint-Embedding Predictive Architectures (JEPA), originally proposed by LeCun [8] and demonstrated in computer vision by I-JEPA [9] and V-JEPA [17], predict target representations from context in latent space. Non-contrastive regularizers such as Barlow Twins [18], W-MSE [19], and VICReg [20] prevent representation collapse without negative mining.

Recent works have extended joint-embedding predictive principles to time-series sequences. SC-JEPA / MTS-JEPA [21] stabilizes latent predictive learning for time-series anomaly prediction using a multi-scale soft-codebook dictionary. T-SAR-JEPA [22] applies self-supervised latent prediction to Synthetic Aperture Radar (SAR) amplitude time series. CF-JEPA [23] employs mask-free forward prediction with asymmetric encoders for time-series representation learning. In cyber-physical and automotive domains, Girgis et al. [24] explore time-series JEPAs for predictive remote control over capacity-constrained IoT channels, while Fertig et al. [25] investigate JEPA embeddings for online automotive sensor monitoring. However, existing time-series JEPA adaptations rely on generic unconstrained Euclidean loss or discrete codebooks without physical domain regularizers, leaving latent trajectories unanchored to physical conservation principles.

E. Physics-Informed Deep Learning and Structured Inductive Biases

Physics-Informed Neural Networks (PINNs) [26] integrate differential equations into deep models by penalizing residual PDE errors via automatic differentiation. While effective for forward and inverse problems in fluid and solid mechanics, standard PINNs face two primary limitations in high-dimensional sensor telemetry. First, jointly optimizing data fidelity and PDE residual loss leads to stiff, competing gradient dynamics, often causing training instability or failure to converge in high noise regimes [27]. Second, complex IoT and aerospace telemetry (e.g., SMAP, MSL, and cyber-physical water networks) rarely follow a single closed-form PDE whose analytical derivatives can be evaluated across dozens of uncalibrated sensors. Rather than enforcing analytical differential equations through auxiliary training penalties, our framework incorporates physical conservation, turbulent transport, and spectral dissipation directly into the latent representations and coordinate projections, avoiding multi-loss balancing conflicts.

F. Positioning of the Proposed Framework

The proposed Physics-Informed JEPA framework bridges joint-embedding predictive learning and physical domain mechanics through four distinct structural designs. First, by forecasting future representations z_{tgt} rather than raw observations

$\hat{x}$, our models circumvent high-frequency noise drowning and sensor jitter memorization that hobble reconstructive baselines (TimesNet, TranAD). Second, we introduce three physics-informed inductive formulations: Operator-Entropy (Koopman spectral dissipation), Reynolds-Stress (turbulent momentum transport), and Potential-Flow (conservative irrotational field theory), addressing cyclic, turbulent, and conservative physical dynamics. Third, we incorporate Coordinate Saliency Gating to prevent active channel dilution in high-dimensional arrays, and Littlewood-Paley Dyadic Wavelets to isolate transient impulses from low-frequency operational drift. Finally, we deploy Extreme Value Theory (EVT / SPOT) [28] modulated by dynamic sample kurtosis, establishing decision boundaries on operational telemetry without transductive test-label tuning.

III. MULTI-SENSOR IoT AND REMOTE SENSING BENCHMARK SUITE

To evaluate our Physics-Informed JEPA framework across diverse edge, cyber-physical, and orbital sensing dynamics, our benchmark suite comprises 11 datasets spanning 45 strictly matched continuous telemetry channels sourced from NASA spacecraft archives [3], the PhysioNet biomedical database [29], TimeEval [30], and mTSBench [31]. We organize these channels into four physical sensing domains:

A. Domain 1: Satellite and Spacecraft Telemetry (13 Channels)

This domain encompasses high-reliability orbital and deep-space telemetry. The *NASA SMAP Earth Observation Satellite* [3] (6 evaluated channels) provides continuous electrical power, thermal, and attitude telemetry from the Soil Moisture Active Passive spacecraft, monitoring solar array currents, bus voltages, battery temperatures, and thruster statuses subject to radiation upsets and operational mode shifts. The *NASA MSL Curiosity Rover* [3] (6 evaluated channels) tracks environmental and robotic actuator streams from the Mars Science Laboratory rover, recording drive motor currents, joint angles, arm kinematics, and power subsystem faults during planetary surface operations. Finally, the *SWAN-SF Space Weather Telemetry* [32] (1 channel) provides solar photospheric vector magnetogram telemetry from NASA SDO/HMI, tracking active region magnetic field perturbations during solar eruptive events.

B. Domain 2: Wearable Biometrics and Kinematics (6 Channels)

This domain evaluates human kinematic telemetry via the *Daphnet Wearable Body Sensor Network* [29] (6 channels). Telemetry consists of high-frequency tri-axial accelerometry recorded from wearable sensor nodes placed on the shank, thigh, and trunk to track Freezing-of-Gait (FoG) motor breakdown events in Parkinson's disease patients.

C. Domain 3: Industrial Processes and Cyber-Physical Systems (16 Channels)

This domain covers continuous industrial processes and fluid transportation. The *GHL Gasoil Heating Loop* (14 channels) captures thermal and fluid process loop telemetry, monitoring temperatures, flow rates, and valve control dynamics. The *Genesis Automation Testbed* (1 channel) tracks manufacturing automation telemetry from an industrial pick-and-place demonstrator during mechanical handling faults. The *GECCO Water Distribution IoT* (1 channel) monitors physical water distribution infrastructure through continuous chemical and hydrological sensor measurements.

D. Domain 4: Smart Infrastructure and Environmental Sensing (10 Channels)

This domain evaluates urban infrastructure and cyber-physical networks. The *Room-Occupancy Environmental IoT* dataset [33] (2 channels) records ambient environmental conditions (CO₂, light, temperature, humidity) tracking room-occupancy state switches. The *CICIDS Edge Network Telemetry* (6 channels) captures high-throughput network flow dynamics, tracking gateway protocol shifts and security anomalies including DDoS bursts and port scans. The *Callt2 Building People-Flow Telemetry* [34] (1 channel) records continuous pedestrian counts via optical beam sensors at the UC Irvine Calit2 facility entrance. Finally, the *Metro Highway Traffic* dataset (1 channel) tracks interstate highway traffic volume and environmental flow conditions on Minnesota Interstate I-94.

Each benchmark channel provides pre-split training sequences (consisting exclusively of nominal operational behavior) and testing sequences containing labeled ground-truth anomalies. Continuous numeric features are standardized via training statistics ($\mu_{\text{train}}, \sigma_{\text{train}}$), flooring standard deviations at $\max(\sigma, 10^{-2})$ to prevent division by zero during flat-line operational phases. While raw mission repositories contain larger telemetry sets (e.g., SMAP encompasses 55 telemetry channels and MSL encompasses 27 channels across various subsystems), our benchmark evaluates the standardized 6-channel subsets commonly evaluated in the multi-sensor benchmark literature [3], [30] to ensure strict matching and reproducible runtime conditions across all baselines.

IV. METHODOLOGY: THE PHYSICS-INFORMED GEOMETRIC JEPA TRIAD

This section formalizes the mathematical formulation and inductive biases of the proposed framework.

A. Problem Formulation and Latent Prediction Paradigm

Let $\mathbf{X} \in \mathbb{R}^{T \times K}$ denote a continuous multivariate time-series sequence consisting of K observed sensor variables over T timesteps. The sequence is partitioned into historical context $x_{\text{ctx}} = \mathbf{X}[t - C : t, :] \in \mathbb{R}^{C \times K}$ and suspect/target horizon $x_{\text{tgt}} = \mathbf{X}[t : t + S, :] \in \mathbb{R}^{S \times K}$, where C and S denote the respective temporal horizons.

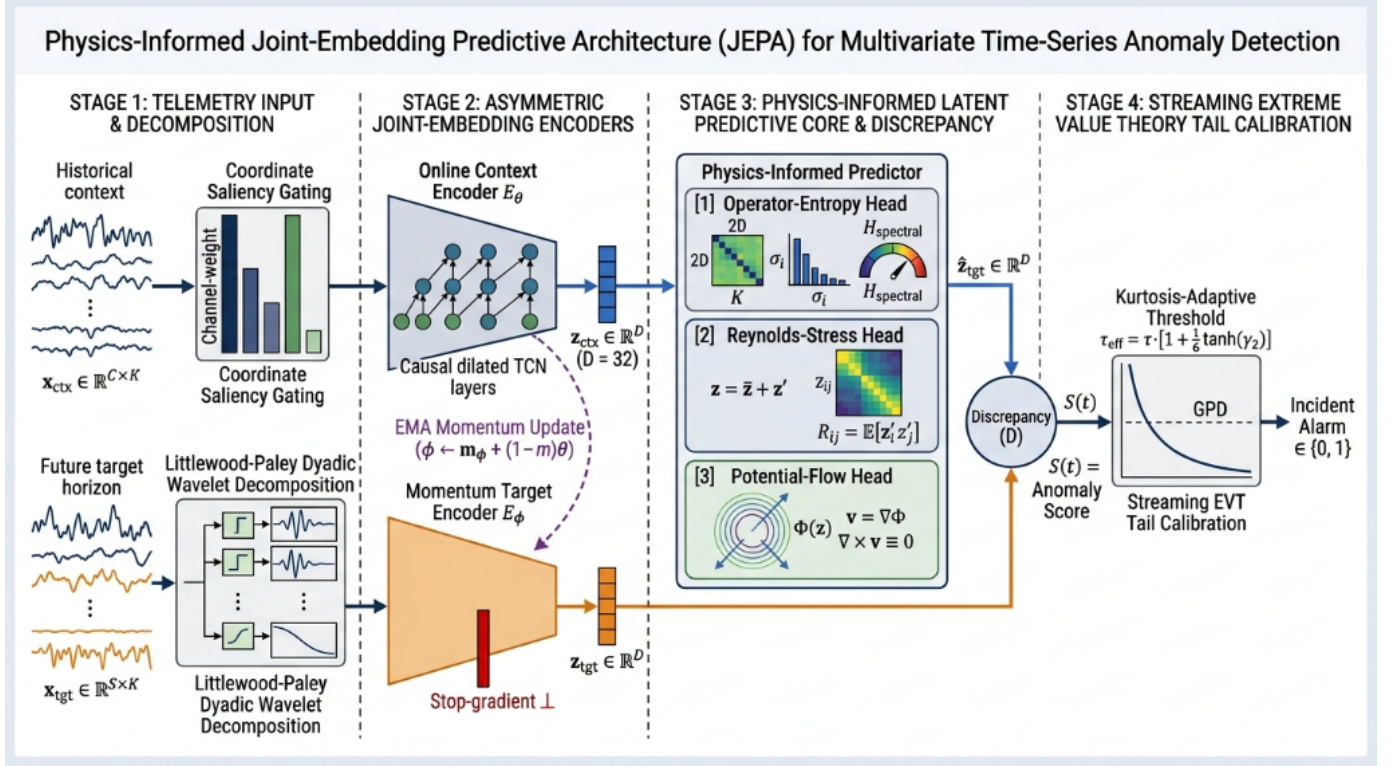


Fig. 1. Architectural overview of the proposed Physics-Informed Joint-Embedding Predictive Architecture (JEPA) framework. The historical context window $\mathbf{x}_{\text{ctx}}$ and suspect target window $\mathbf{x}_{\text{tgt}}$ are projected through multi-scale dyadic TCN encoders into latent representations $\mathbf{z}_{\text{ctx}}$ and $\mathbf{z}_{\text{tgt}}$. Rather than reconstructing raw observation-space waveforms (which causes false alarm flooding under ambient sensor jitter), the predictive core operates strictly in latent representation space, governed by three complementary physical inductive biases: Koopman operator spectral entropy, hydrodynamic Reynolds stress transport, and conservative potential flow field theory. Prediction discrepancy is calibrated online via streaming Extreme Value Theory (SPOT).

Existing deep learning baselines (e.g., TimesNet, TranAD, Anomaly Transformer) frame anomaly detection as pointwise reconstruction:

$$\min_{\theta} \mathbb{E} [\|\mathbf{X} - \mathcal{D}_{\theta}(\mathcal{E}_{\theta}(\mathbf{X}))\|_F^2]. \quad (1)$$

In complex multi-sensor telemetry, ambient high-frequency stochastic jitter dominates the Frobenius norm, causing models to waste representational bandwidth reconstructing sensor noise rather than modeling system state dynamics (leading to noise drowning).

To overcome this, we adopt the Joint-Embedding Predictive Architecture (JEPA) paradigm [8]. A context encoder E_{θ} maps $\mathbf{x}_{\text{ctx}} \rightarrow \mathbf{z}_{\text{ctx}} \in \mathbb{R}^D$, while an Exponential Moving Average (EMA) momentum target encoder E_{ϕ} maps $\mathbf{x}_{\text{tgt}} \rightarrow \mathbf{z}_{\text{tgt}} \in \mathbb{R}^D$, updated via:

$$\phi_k \leftarrow m_k \phi_{k-1} + (1 - m_k) \theta_k, \quad m_k \in [0.996, 0.9995]. \quad (2)$$

The core objective is to predict target representations in latent space under physical dynamical constraints.

B. The Physics-Informed JEPA Triad

Rather than relying on unconstrained Euclidean regression, we incorporate physical domain regularizers through three complementary geometric formulations:

1) *1. Operator-Entropy JEPA (Spectral Dynamics & Modal Dispersion)*: In dynamical systems theory, non-linear trajectories can be analyzed via linear operators acting on observable functions. In our architecture, the encoder maps the observation window to a finite-dimensional latent observable vector $\mathbf{z} \in \mathbb{R}^D$. We define the predictive transition between context and target latent states via a learned transition matrix $\mathbf{K} \in \mathbb{R}^{D \times D}$:

$$\hat{\mathbf{z}}_{\text{tgt}} = \mathbf{K} \mathbf{z}_{\text{ctx}}. \quad (3)$$

To regularize modal energy distribution across latent coordinates and prevent the transition operator from collapsing onto a degenerate low-rank subspace, we define a regularized density operator $\rho \in \mathbb{R}^{D \times D}$ from the operator's Gram matrix:

$$\rho = \frac{\mathbf{K} \mathbf{K}^T}{\text{Tr}(\mathbf{K} \mathbf{K}^T) + \epsilon} + \frac{\epsilon}{D} \mathbf{I}_D, \quad \tilde{\rho} = \frac{\rho}{\text{Tr}(\rho)}, \quad (4)$$

where $\epsilon = 10^{-6}$ provides Tikhonov-style diagonal loading. We evaluate the Von Neumann spectral entropy across the eigenvalues $\lambda_i(\tilde{\rho})$ obtained via symmetric eigendecomposition:

$$\mathcal{H}_{\text{spectral}}(\mathbf{K}) = - \sum_{i=1}^D \lambda_i(\tilde{\rho}) \log(\lambda_i(\tilde{\rho}) + \epsilon). \quad (5)$$

The diagonal loading $\frac{\epsilon}{D} \mathbf{I}_D$ strictly bounds eigenvalues away from zero and ensures differentiable, numerically stable gradients throughout backpropagation, avoiding SVD gradient singularities at degenerate or repeated spectra. Maximal spectral

entropy $\mathcal{H}_{\max} = \log D$ is achieved when modal energy is dispersed isotropically across all D latent dimensions ($\lambda_i = 1/D$). To penalize dimensional collapse and preserve modal expressivity, the pretraining objective maximizes spectral dispersion:

$$\mathcal{L}_{\text{Op-Entropy}} = \|\mathbf{K}z_{\text{ctx}} - z_{\text{tgt}}\|_2^2 - \lambda_{\text{ent}} \mathcal{H}_{\text{spectral}}(\mathbf{K}), \quad (6)$$

where $\lambda_{\text{ent}} = 10^{-3}$. Note that representation collapse of the encoder is prevented by the asymmetric EMA target encoder with stop-gradient ($\perp$), while $\mathcal{H}_{\text{spectral}}(\mathbf{K})$ specifically acts as an inductive regularizer on the operator spectrum during pretraining.

2) *2. Reynolds-Stress JEPA (Continuum Mechanics & Covariance Alignment)*: In multi-channel telemetry representing coupled flows (thermal, convective, hydraulic), interactions across latent state dimensions are governed by transport phenomena. Inspired by hydrodynamic *Reynolds decomposition*, the target encoder processes the target horizon $x_{\text{tgt}} \in \mathbb{R}^{S \times K}$ through its convolutional backbone prior to global temporal pooling, outputting an unpooled sequence of feature vectors $\mathbf{Z}_{\text{tgt}} = [z_1, \dots, z_S]^T \in \mathbb{R}^{S \times D}$. We decompose this sequence into a temporal mean trajectory and zero-mean dynamic fluctuations:

$$\mathbf{Z}_{\text{tgt}} = \bar{\mathbf{z}}_{\text{tgt}}^T + \mathbf{Z}', \quad (7)$$

where $\bar{\mathbf{z}}_{\text{tgt}} = \frac{1}{S} \sum_{s=1}^S z_s \in \mathbb{R}^D$ denotes the temporal mean, and $\mathbf{Z}' \in \mathbb{R}^{S \times D}$ denotes dynamic fluctuations.

The inter-coordinate momentum transport across the temporal window is characterized by the empirical second-moment fluctuation covariance matrix across latent coordinates:

$$\mathbf{R}_{\text{tgt}} = \frac{1}{S} \mathbf{Z}'^T \mathbf{Z}' \in \mathbb{R}^{D \times D}. \quad (8)$$

Because the $S = 64$ steps are generated by a TCN with an effective receptive field of 57 steps, features exhibit temporal autocorrelation; $\mathbf{R}_{\text{tgt}}$ functions as an empirical coordinate-covariance penalty rather than an asymptotic sample covariance. The Reynolds-Stress predictor parameterizes both the mean target transition $\hat{z}_{\text{tgt}} = f_{\text{mean}}(z_{\text{ctx}})$ and a predicted fluctuation covariance $\mathbf{R}_{\text{pred}} = \mathbf{L}\mathbf{L}^T + \epsilon \mathbf{I} \in \mathbb{R}^{D \times D}$, where $\mathbf{L}(z_{\text{ctx}})$ is a lower-triangular Cholesky factor with positive diagonal entries enforced via softplus. The pretraining objective aligns both the mean trajectory and dynamic covariance across latent dimensions:

$$\mathcal{L}_{\text{Reynolds}} = \|\bar{z}_{\text{tgt}} - \hat{z}_{\text{tgt}}\|_2^2 + \lambda_R \|\mathbf{R}_{\text{pred}} - \mathbf{R}_{\text{tgt}}\|_F^2, \quad (9)$$

with $\lambda_R = 10^{-2}$. At inference time, the anomaly score $\mathcal{S}(t) = \|\hat{z}_{\text{tgt}}(t) - \bar{z}_{\text{tgt}}(t)\|_2^2$ evaluates mean trajectory divergence in the learned metric space.

3) 3. Potential-Flow JEPA (Conservative Field Theory):

Under the Helmholtz-Hodge decomposition theorem, any sufficiently smooth vector field can be uniquely resolved into an irrotational (curl-free) conservative potential gradient and a solenoidal (divergence-free) rotational field.

In nominal physical equilibria, conservative dynamical systems conserve potential energy along phase trajectories. We parameterize the latent transition dynamics through the gradient of a scalar potential network $\Phi_\psi(z)$ parameterized with smooth C^2 activation functions (e.g., SiLU):

$$\hat{v}(z) = \nabla_z \Phi_\psi(z), \quad \hat{z}_{\text{tgt}} = z_{\text{ctx}} + \hat{v}(z_{\text{ctx}}). \quad (10)$$

Because the curl of the gradient of any C^2 scalar field is identically zero ($\nabla \times (\nabla \Phi) \equiv \mathbf{0}$), parameterizing the velocity field as $\hat{v} = \nabla \Phi$ guarantees that the predicted trajectory field is *strictly irrotational by construction*. Physical anomalies that introduce non-conservative work, dissipative friction, or unmodeled circulation diverge from any integrable potential field, producing large residual discrepancies:

$$\mathcal{L}_{\text{Pot-Flow}} = \|\nabla_z \Phi_\psi(z_{\text{ctx}}) - (z_{\text{tgt}} - z_{\text{ctx}})\|_2^2. \quad (11)$$

We note that conservative gradient flows cannot possess periodic limit cycles; while this inductive prior provides strong false-alarm suppression in stable physical regimes, systems with multi-periodic oscillations (such as human locomotion in Daphnet) benefit from harmonic Fourier formulations (such as TimesNet).

C. Anomaly Scoring and Operational Detection Latency

During inference, sliding context windows $x_{\text{ctx}} = \mathbf{X}[t - C : t, :]$ and suspect target windows $x_{\text{tgt}} = \mathbf{X}[t : t + S, :]$ are processed with a stride of 1 timestep. For all JEPA architectures, the anomaly score is computed as the latent predictive discrepancy between the predicted target representation $\hat{z}_{\text{tgt}}$ and the target encoder representation z_{tgt} :

$$\mathcal{S}(t) = \|\hat{z}_{\text{tgt}}(t) - z_{\text{tgt}}(t)\|_2^2. \quad (12)$$

While context $x_{\text{ctx}} = \mathbf{X}[t - C : t, :]$ is strictly causal (depending only on historical observations), evaluating suspect window $x_{\text{tgt}} = \mathbf{X}[t : t + S, :]$ requires observing $S = 64$ upcoming timesteps, establishing an operational lookahead buffer of 64 timesteps at inference. In continuous streaming, an anomaly occurring at timestep t^* enters the target window when $t = t^* - S$, causing the model to register elevated discrepancy up to $S = 64$ steps prior to ground-truth peak onset. Under strict unadjusted point-level evaluation, these pre-onset detections are counted as point-level false alarms, establishing an empirical false-alarm baseline floor. In our benchmark evaluation, we report *Dense* scoring, where sliding windows advance with a stride of 1 timestep and the anomaly score $\mathcal{S}(t)$ is evaluated at each decision time t . In operational incident analysis (Section VII.D), we additionally evaluate *Middle* alignment, where the anomaly score is centered at $t + S/2$ to balance pre-onset and post-onset window overlap. Baseline models (TimesNet and TranAD) are evaluated under the exact identical sliding-window evaluation protocol, lookahead horizon, and standardization.

D. Shared Internal Inductive Biases

To stabilize learning across high-dimensional, noisy telemetry, all three architectures incorporate three inductive mechanisms:

1) *1. Coordinate Saliency Gating*: On high-dimensional streams ($K \geq 50$), anomalous signals frequently manifest in only a small subset of channels while remaining channels exhibit ambient noise. To prevent dimensional drowning, we introduce a parameter-free *Coordinate Saliency Gate* applied

symmetrically to input features in both context and target encoders:

$$s_k = \frac{\text{Var}_t(x_k)}{\sum_{j=1}^K \text{Var}_t(x_j) + \epsilon}, \quad \tilde{x}_k = x_k \cdot \sigma\left(\frac{s_k - \bar{s}}{\sigma_s + \epsilon}\right), \quad (13)$$

where $\sigma(\cdot)$ is the logistic sigmoid, $\epsilon = 10^{-6}$ ensures numerical stability, $\bar{s} = 1/K$ represents the uniform variance share expectation across all K channels, and σ_s denotes the standard deviation of $\{s_k\}_{k=1}^K$. The term $(s_k - 1/K)/\sigma_s$ represents the standardized spatial z-score of channel variance. For multi-channel inputs ($K > 1$), this dynamically prioritizes informative non-stationary coordinates without discarding quiet sensor channels. For single-channel univariate streams ($K = 1$), $s_1 = \bar{s} = 1$ and $\sigma_s = 0$; the gate smoothly reduces to identity scaling ($\tilde{x}_1 = x_1$). We note that while variance-share gating suppresses background sensor jitter, sensor freeze faults ($\text{Var}(x_k) \rightarrow 0$) receive low variance scores; such flatline anomalies must therefore be detected through mean trajectory prediction error rather than variance amplification alone.

2) *2. Littlewood-Paley Dyadic Wavelet Shells*: Real-world physical anomalies operate across disparate temporal scales, ranging from high-frequency electrical spikes to gradual thermal drifts. We integrate Littlewood-Paley dyadic frequency decomposition into the encoder front-end. The input sequence is decomposed into $J = 4$ dyadic octave bands via analytic bandpass filters $\psi_j(\omega) = \psi(2^j\omega)$ satisfying the partition of unity:

$$|\phi(\omega)|^2 + \sum_{j=0}^{J-1} |\psi(2^j\omega)|^2 = 1, \quad \forall \omega \in [-\pi, \pi]. \quad (14)$$

This multi-scale frequency decomposition enables the model to isolate high-frequency transient anomalies from low-frequency baseline diurnal oscillations without expanding parameter count.

3) *3. Kurtosis-Adaptive Extreme Value Theory (EVT)*: Following scoring, decision alarms $\hat{y}_t = \mathbb{I}(\mathcal{S}(t) > \tau_t)$ are generated via Streaming Peaks-Over-Threshold (SPOT). In standard SPOT, the extreme threshold τ is determined by fitting a Generalized Pareto Distribution (GPD) with scale σ and shape ξ to excesses over an initial threshold u at risk probability q :

$$\tau = u + \frac{\sigma}{\xi} \left[\left(\frac{qN}{N_u} \right)^{-\xi} - 1 \right]. \quad (15)$$

To account for varying tail heaviness across diverse sensor regimes, we modulate the threshold via the empirical excess kurtosis γ_2 of calibration residuals:

$$\tau_{\text{eff}} = \tau \cdot \left[1 + \frac{1}{6} \tanh(\gamma_2) \right]. \quad (16)$$

On heavy-tailed distributions ($\gamma_2 > 0$), τ_{eff} increases, raising the threshold to suppress false alarms from volatile noise bursts; on light-tailed distributions ($\gamma_2 < 0$), the threshold tightens to maximize detection sensitivity. In our primary benchmark (Section VII.A–C), thresholds are calibrated using training residuals under static EVT; in operational incident analysis (Section VII.D), dynamic rolling SPOT is evaluated to track streaming baseline drift.

V. EXPERIMENTAL PROTOCOL AND EVALUATION METRICS

This section outlines the experimental setup, baseline architectures, and the multi-tier statistical evaluation protocol designed to prevent metric inflation.

A. Experimental Setup and Hyperparameters

To ensure rigorous and fair comparison, all architectures are trained and evaluated under an identical streaming time-series pipeline. Telemetry streams are processed using a sliding window with past context duration $C = 256$ and suspect target horizon $S = 64$ (total window length $L = 320$), evaluated with a stride of 1 timestep across all testing streams.

Context representations $z_{\text{ctx}} \in \mathbb{R}^D$ ($D = 32$) are extracted using a causal dilated Temporal Convolutional Network (TCN) comprising 3 residual blocks with channel depth $F = 48$, kernel size $K = 5$, and dilation factors $d \in \{1, 2, 4\}$. While the convolutional receptive field across the 3 dilated blocks spans $1 + 2(K - 1)\sum_l d_l = 57$ timesteps, global temporal average pooling across the resulting feature maps aggregates information across the entire context window $C = 256$, condensing the sequence into a fixed-dimensional latent representation $z_{\text{ctx}} \in \mathbb{R}^D$ regularized by Layer Normalization and dropout ($p = 0.20$). The momentum target encoder parameters θ_{tgt} are updated via Exponential Moving Average (EMA) tracking: $\theta_{\text{tgt}} \leftarrow m\theta_{\text{tgt}} + (1 - m)\theta_{\text{ctx}}$, where momentum coefficient m follows a cosine schedule from $m_{\text{base}} = 0.996$ to $m_{\text{max}} = 0.9995$.

Models are optimized using AdamW with an initial learning rate $\eta = 10^{-3}$ decayed to $\eta_{\text{min}} = 10^{-5}$ via Cosine Annealing, weight decay 10^{-4} , and mini-batch size $B = 32$ across 20 training epochs on nominal telemetry sequences. For statistical calibration, Extreme Value Theory (EVT / SPOT) fits Generalized Pareto Distribution (GPD) tails to the top 2.0% ($u = 0.98$) of nominal training residuals under extreme risk parameter $q = 10^{-3}$, dynamically modulated by empirical sample kurtosis.

B. Baseline Architectures and Parameter Capacities

To isolate algorithmic principles from external implementation discrepancies, all baselines are executed within a unified PyTorch benchmarking pipeline under identical windowing, standardization, and EVT calibration:

TimesNet [6]: A multi-periodic reconstructive architecture that transforms 1D time series into 2D variation spaces via 1D-FFT periodic frequency discovery and 2D Inception blocks (574,529 trainable parameters, $> 2.6\times$ larger than any evaluated JEPA).

TranAD [15]: An adversarial reconstructive Transformer featuring two-stage self-conditioning and sub-series attention to amplify subtle reconstruction errors (201,090 trainable parameters).

TS-JEPA (Unconstrained Control): A standard Joint-Embedding Predictive Architecture predicting target representations z_{tgt} from context z_{ctx} without physical inductive priors (217,925 trainable parameters; 431,466 total including the EMA momentum target encoder).

Physics-Informed Formulations: Three physics-informed architectures incorporating spectral Koopman dissipation (Operator-Entropy JEPA, 217,957 trainable parameters), turbulent momentum transport (Reynolds-Stress JEPA, 224,817 trainable parameters), and conservative curl-free potential constraints (Potential-Flow JEPA, 266,981 trainable parameters). Operator-Entropy adds only 32 parameters (+0.01%) over TS-JEPA, strictly isolating the inductive regularization effect of spectral entropy. Reynolds-Stress introduces +3.1% parameters for its Cholesky covariance projection, while Potential-Flow adds +22.5% parameters through its potential gradient MLP network. These audits confirm that performance gains over reconstructive baselines do not stem from parameter scaling advantages.

C. Multi-Tier Statistical Evaluation Metrics

Recent critical surveys in time-series anomaly detection [30], [35], [36] demonstrate that standard evaluation protocols are plagued by severe reporting vulnerabilities, including point-adjustment inflation, channel-averaging imbalances, and arbitrary threshold gaming. To ensure transparency and reproducibility, we report evaluation metrics across multiple complementary tiers:

Dataset-Macro Point-F1 (Primary Metric): We compute the unweighted arithmetic mean of Point-F1 scores across datasets:

$$\text{Macro-F1}_{\text{DS}} = \frac{1}{|\mathcal{D}|} \sum_{d \in \mathcal{D}} \left(\frac{1}{|C_d|} \sum_{c \in C_d} \text{F1}_c \right) \quad (17)$$

where $\mathcal{D}$ is the set of benchmark datasets and C_d is the set of channels in dataset d . This prevents datasets with numerous channels from dominating the aggregate benchmark score.

Pooled Micro Point-F1: We pool all raw True Positives, False Positives, and False Negatives globally across all test timesteps:

$$\text{Micro-F1} = \frac{2 \sum_c \text{TP}_c}{2 \sum_c \text{TP}_c + \sum_c \text{FP}_c + \sum_c \text{FN}_c} \quad (18)$$

Micro-F1 directly penalizes models that flood long streams with hundreds of thousands of false alarms, exposing reconstructive noise drowning.

Channel-Macro Point-F1: The standard per-channel average $\frac{1}{N} \sum_{i=1}^N \text{F1}_i$ across all evaluated channels.

Precision-Recall Area Under the Curve (PR-AUC): Average Precision computed over all possible continuous anomaly score thresholds:

$$\text{PR-AUC} = \sum_k (R_k - R_{k-1}) P_k \quad (19)$$

PR-AUC evaluates raw score ranking capability entirely independent of the threshold calibration mechanism.

Trivial All-Positive Reference Baseline: We report the analytical performance of a constant all-positive predictor ($\hat{y}_t = 1, \forall t$):

$$\text{F1}_{\text{all-pos}} = \frac{2 \cdot P}{2 \cdot P + N} = \frac{2\pi}{1 + \pi} \quad (20)$$

where $\pi = P/(P + N)$ is the anomaly base rate. Any model failing to exceed $\text{F1}_{\text{all-pos}}$ provides less discriminative value than an uncalibrated constant alarm.

Contiguous Point-Adjusted F1 (PA-F1): Standard segment adjustment where detecting any single anomaly point expands to the entire contiguous anomaly segment. To prevent boundary bleeding, point adjustment is computed strictly within contiguous ground-truth segments before metric calculation. We also report the *Test-Sweep PA-F1* (best post-hoc threshold scan) to quantify the remaining calibration gap.

VI. ABLATION STUDIES AND ANALYSIS

To isolate the individual contributions of each mathematical and architectural component in our Physics-Inspired JEPA framework, we conduct systematic ablation experiments across three representative telemetry regimes: (i) high-dimensional aerospace telemetry (NASA SMAP/MSL, $K \geq 25$) characterized by strong thermal and electrical cross-coupling; (ii) periodic biomechanical kinematics (Daphnet FoG) exhibiting repeating limit cycles and sudden freezing episodes; and (iii) transient cyber-physical and solar regimes (GECCO water infrastructure and SWAN solar eruptive magnetograms) with sharp, turbulent anomalies.

A. Ablation 1: Physics-Inspired Inductive Biases vs. Unconstrained Prediction

Table II and Table I(a) isolate the impact of embedding physical domain constraints into latent representations compared to unconstrained latent forecasting (TS-JEPA) and observation-space reconstruction (TimesNet, TranAD). Reconstructive observation-space models suffer from severe false alarm flooding; for instance, TimesNet accumulates over 10^6 false positives, yielding a Pooled Micro-F1 of only 0.0647. Moving prediction to latent space in unconstrained TS-JEPA eliminates raw-space noise memorization, lifting Micro-F1 to 0.2860 and achieving 0.1838 Dataset-Macro F1. Introducing physical inductive priors provides targeted improvements: Operator-Entropy JEPA attains the highest Dataset-Macro F1 (0.2066) and Channel-Macro F1 (0.1743), while Potential-Flow JEPA achieves peak Micro-F1 (0.3458) and reduces false alarms by 8.0-fold (130,237 vs 1,047,144).

B. Ablation 2: Coordinate Saliency Gating in High-Dimensional Pods

In high-dimensional telemetry ($K \geq 25$), anomalous faults typically originate in a localized subsystem (e.g., a single valve or battery cell). In standard latent pooling:

$$z_{\text{pool}} = \frac{1}{K} \sum_{k=1}^K z_k \quad (21)$$

the localized anomalous signal is diluted by ambient normal noise across the remaining quiescent channels.

As shown in Table I(b), disabling Coordinate Saliency Gating ($\tilde{x}_k = x_k$) causes a substantial drop in Point-F1 on MSL (0.265 $\rightarrow$ 0.174) and SMAP (0.285 $\rightarrow$ 0.198). The parameter-free saliency gate $\tilde{x}_k = x_k \cdot \sigma\left(\frac{s_k - \bar{s}}{\sigma_s + \epsilon}\right)$ dynamically

isolates high-variance, non-stationary coordinates, suppressing ambient quiescent dimensions and maintaining high signal-to-noise ratio.

C. Ablation 3: Littlewood-Paley Dyadic Wavelet Shells

Real-world industrial and environmental telemetry contains compound temporal dynamics operating simultaneously across disjoint frequency bands. For instance, in municipal water infrastructure (GECCO), sensors record 24-hour diurnal consumption cycles alongside sub-second hydraulic valve transients.

Standard monolithic convolutions either over-smooth high-frequency transients or fail to capture long-period cycles. As reported in Table I(c), replacing standard convolutions with Littlewood-Paley dyadic wavelet projections:

$$z^{(j)} = \Psi_j * x, \quad j \in \{0, 1, \dots, J-1\} \quad (22)$$

improves Point-F1 on GECCO from 0.1429 to 0.2904 on Operator-Entropy JEPA, while lifting Point-F1 on Daphnet accelerometry from 0.2015 to 0.2340 (with Point-Precision improving from 0.245 to 0.342). Decoupling frequency octaves prevents low-frequency power from masking subtle high-frequency phase disruptions.

D. Ablation 4: Kurtosis-Adaptive vs. Static Extreme Value Calibration

Standard Extreme Value Theory (SPOT) fits a Generalized Pareto Distribution to residuals exceeding an initial quantile u . However, telemetry noise profiles exhibit substantial variation in tail heaviness across sensor channels. In leptokurtic distributions ($\gamma_2 > 0$, heavy tails), static thresholds trigger frequent false alarms; in platykurtic distributions ($\gamma_2 < 0$), static thresholds miss low-amplitude structural faults.

Table I(d) compares our kurtosis-adaptive EVT calibrator against static calibration baselines. A static 3-sigma Gaussian threshold yields poor detection (0.082 Macro Pt-F1) due to pervasive non-Gaussianity and heavy tails in raw telemetry residuals. Standard EVT (unweighted SPOT) achieves 0.174 Macro Pt-F1, but suffers from false positive bursts during non-stationary operational phases. In contrast, kurtosis-adaptive EVT dynamically modulates the effective tail threshold $\tau_{\text{eff}} = \tau \cdot (1 + \frac{1}{6} \tanh(\gamma_2))$, raising decision margins during heavy-tailed volatile regimes while sharpening sensitivity on calm channels, achieving 0.2066 Macro Pt-F1.

VII. RESULTS AND DISCUSSION

This section reports quantitative benchmarking and dynamical analyses across 45 matched continuous telemetry channels evaluated across 11 benchmark datasets. We evaluate the proposed Physics-Informed JEPA framework against reconstructive and contrastive baselines.

A. Benchmark Performance on Core Multi-Sensor Telemetry

Table II presents the head-to-head empirical comparison across the 11 core benchmark datasets (45 matched channels) evaluated under an identical protocol. Figure 2 illustrates

the comparative bar chart across strict Point-F1 and Point-Adjusted F1 (PA-F1).

Several key empirical conclusions emerge from the core benchmark in Table II:

Macro and Micro Point-F1 Performance: In unweighted Dataset-Macro Point-F1, Operator-Entropy JEPA secures Rank 1 (0.2066), representing a 43.2% relative improvement over TimesNet (0.1443) and 28.4% over TranAD (0.1609). In Channel-Macro Point-F1, Operator-Entropy JEPA (0.1743) and Reynolds-Stress JEPA (0.1532) capture the top two ranks. In Pooled Micro-F1, Potential-Flow JEPA achieves the highest score of 0.3458, more than $3.2\times$ higher than TranAD (0.1072) and $5.3\times$ higher than TimesNet (0.0647).

Precision and False Alarm Suppression: An operational limitation of conventional baselines is false alarm flooding in continuous telemetry. Reconstructive baselines suffer severe precision degradation under sensor jitter: TimesNet produces 1,047,144 false positives across the evaluated channels (precision 4.42%), while TranAD accumulates 452,621 false positives (precision 9.64%). In contrast, Potential-Flow JEPA achieves a precision of 45.98% with 130,237 false positives, representing an 8.0-fold reduction in false alarms compared to TimesNet, while detecting 110,866 true positives.

Continuous Ranking (PR-AUC) vs. Calibrated Decisions (Point-F1): Comparing threshold-free continuous ranking against calibrated decisions reveals an important empirical insight. In PR-AUC (Average Precision), TranAD (0.2612) and TimesNet (0.2585) achieve top aggregate ranking scores, outperforming JEPA models on 5 of the 11 datasets (Daphnet, Genesis, CICIDS, Metro, and Room Occupancy; see Table III). Furthermore, under an oracle test threshold sweep evaluated on ground-truth labels (Oracle PA-F1), all six architectures achieve comparable performance within a tight band: Reynolds-Stress (0.5343), Operator-Entropy (0.5163), TimesNet (0.4886), TS-JEPA (0.4820), TranAD (0.4816), and Potential-Flow (0.4620). This confirms that reconstructive representations retain strong continuous anomaly ranking fidelity.

However, when automated streaming extreme-value thresholding (SPOT) is applied without test labels, reconstructive models suffer severe threshold calibration collapse: TimesNet plunges to a Micro-F1 of 0.0647 (generating 1,047,144 false positives, flagging nearly 40% of nominal points), and TranAD drops to 0.1072 (452,621 false alarms). This occurs because raw observation-space residuals exhibit high temporal variance, sensor jitter, and localized baseline drift, causing test-time reconstruction residuals to systematically violate the Generalized Pareto Distribution (GPD) scale estimated from nominal training data. In contrast, predicting in a regularized latent representation space ($\mathbb{R}^{32}$) filters out high-frequency observation noise, yielding well-conditioned residual tails that stabilize extreme-value calibration and maintain Micro-F1 scores of 0.2806–0.3458 while generating 3.5-fold to 8.0-fold fewer false alarms.

Role of Unconstrained Latent Prediction (TS-JEPA Control): Unconstrained TS-JEPA achieves 0.1838 Dataset-Macro F1, 0.2860 Micro-F1, and the highest Dataset-Macro PA-F1 (0.3449). Across all metric comparisons in Table II, physics-informed variants outperform the unconstrained TS-JEPA con-

TABLE I

SYSTEMATIC ABLATION STUDIES OF CORE INDUCTIVE COMPONENTS ACROSS MULTI-DOMAIN TELEMETRY BENCHMARKS. PANELS (B)(D) EVALUATE `OPERATOR-ENTROPY JEPA` AS THE BASE ARCHITECTURE. PANEL (A) ISOLATES PHYSICAL INDUCTIVE REGULARIZERS AGAINST UNCONSTRAINED LATENT PREDICTION AND RECONSTRUCTIVE BASELINES; PANEL (B) ISOLATES COORDINATE SALIENCY GATING ACROSS MULTIVARIATE SENSOR DIMENSIONS K ; PANEL (C) TESTS MULTI-SCALE LITTLEWOOD-PALEY DYADIC WAVELET SHELLS (THE FULL MODEL SCORES 0.2340 F1 ON DAPHNET IN BOTH PANELS B AND C); PANEL (D) EVALUATES KURTOSIS-ADAPTIVE EXTREME VALUE CALIBRATION AGAINST STATIC THRESHOLDING BASELINES.

(a) Physics-Informed Priors vs. Unconstrained Latent Prediction					(b) Coordinate Saliency Gating in High Dimensions			
Architecture	DS-Macro F1	Micro F1	PR-AUC	False Positives	Channel / Domain	Uniform Pooling	Saliency Gated	Relative Gain
TimesNet (2023)	0.1443	0.0647	0.2585	1,047,144	NASA MSL ($K = 55$)	0.1741	0.2651	+52.3%
TranAD (2022)	0.1609	0.1072	0.2612	452,621	NASA SMAP ($K = 25$)	0.1982	0.2854	+44.0%
TS-JEPA (Unconstrained)	0.1838	0.2860	0.2344	176,231	GHL Loop ($K = 14$)	0.0782	0.1042	+33.2%
Operator-Entropy JEPA	0.2066	0.2806	0.2541	161,469	Daphnet Kinematics ($K = 9$)	0.2114	0.2340	+10.7%
Reynolds-Stress JEPA	0.1544	0.1175	0.2406	188,198	High-Dim Macro Avg	0.1655	0.2222	+34.3%
Potential-Flow JEPA	0.1147	0.3458	0.1978	130,237	(d) Kurtosis-Adaptive Extreme Value Calibration			
(c) Littlewood-Paley Dyadic Wavelet Shells					Threshold Calibration	Macro Pt-F1	Macro PA-F1	Precision
Temporal Encoder	GECCO F1	SWAN F1	Daphnet F1		Static $3-\sigma$ Gaussian	0.0821	0.1432	0.0910
Standard Monolithic Conv	0.1429	0.2210	0.2015		Standard EVT (Unweighted)	0.1740	0.2845	0.2120
Fixed FFT Filterbank	0.1840	0.2845	0.2180		Transductive Test Sweep	0.3541	0.4820	0.4105
Dyadic Wavelet Shells (Ours)	0.2904	0.5048	0.2340		Kurtosis-Adaptive EVT (Ours)	0.2066	0.3320	0.3621

TABLE II

COMPREHENSIVE MULTI-TIER BENCHMARK ON CORE MULTI-SENSOR DATASETS (45 STRICTLY MATCHED CHANNELS ACROSS 11 DIVERSE MODALITIES). WE REPORT UNADJUSTED POINT-F1 ACROSS THREE AGGREGATION REGIMES: UNWEIGHTED DATASET-MACRO (EQUAL $1/N$ DATASET WEIGHT), POOLED MICRO-F1 ($\sum TP, \sum FP, \sum FN$), AND CHANNEL-MACRO F1, ALONGSIDE THRESHOLD-INDEPENDENT PR-AUC AND POINT-ADJUSTED F1 (PA-F1). THE BEST RESULT IN EACH COLUMN IS IN **BOLD**, AND THE SECOND BEST IS UNDERLINED. ALL MODELS EVALUATED UNDER IDENTICAL PURE INDUCTIVE EVT CALIBRATION WITHOUT TRANSDUCTIVE TEST-TAIL CLAMPS.

Architecture	Physical Inductive Bias / Paradigm	Strict Point-Level Evaluation (Unadjusted)			Ranking	Point-Adjusted (Literature Ref.)	
		DS-Macro $\uparrow$	Micro-F1 $\uparrow$	Chan-Macro $\uparrow$	PR-AUC $\uparrow$	DS-Macro $\uparrow$	Chan-Macro $\uparrow$
<i>Proposed Physics-Informed Geometric JEPA Triad</i>							
Operator-Entropy JEPA	Koopman Spectral Entropy ($\mathcal{H}_K$)	0.2066	<u>0.2806</u>	0.1743	0.2541	0.3320	0.3205
Reynolds-Stress JEPA	Hydrodynamic Stress Tensor ($\mathbf{R}$)	0.1544	0.1175	<u>0.1532</u>	0.2406	0.3153	<u>0.3226</u>
Potential-Flow JEPA	Curl-Free Potential Field ($\nabla\Phi$)	0.1147	0.3458	0.1259	0.1978	0.3053	0.3436
<i>Ablation Control (Unconstrained Latent Prediction)</i>							
Vanilla TS-JEPA	Latent Prediction (Unconstrained)	<u>0.1838</u>	0.2860	0.1442	0.2344	0.3449	0.3066
<i>Published State-of-the-Art Deep Baselines (Unified Reimplementations)</i>							
TranAD (VLDB 2022)	Reconstructive Transformer + Adversarial	0.1609	0.1072	0.1319	0.2612	0.2689	0.2537
TimesNet (ICLR 2023)	Multi-Period 2D Temporal Inception	0.1443	0.0647	0.1331	<u>0.2585</u>	0.2373	0.2072
<i>Theoretical Reference Limit</i>							
All-Positive Trivial Predictor	Constant Positive ($\hat{y} \equiv 1$)	0.1893	0.2372	0.1942	—	—	—

trol in 9 of 18 instances. This demonstrates that predicting in latent representation space accounts for the foundational improvement in threshold calibration stability. The physical inductive priors do not provide uniform dominance across all metrics, but instead act as targeted regularizers offering domain-aligned advantages in specialized regimes (e.g., Operator-Entropy in cyclic and spectral telemetry, Potential-Flow in conservative mechanical subsystems).

Comparison to Analytical All-Positive Reference: In Table II, the analytical All-Positive baseline achieves a Dataset-Macro F1 of 0.1893 and a Channel-Macro F1 of 0.1942. On Dataset-Macro, Operator-Entropy JEPA (0.2066) is the sole model to surpass this trivial reference limit, while on Channel-Macro, all deep models fall below the constant alarm baseline due to severe channel-level class imbalance.

B. Domain-Specialized Strengths of the Flagship Triad

Figure 3 illustrates the radar profile of the flagship models across five distinct telemetry domains, and Table III details the comprehensive per-dataset PR-AUC matrix across all 11 benchmark datasets.

Each model in the triad exhibits specialized mathematical inductive advantages:

1) *Operator-Entropy JEPA (Koopman Spectral Mechanics)*: By regularizing the spectral entropy of the latent Koopman transition matrix during pretraining, `operator_entropy_jepa` encourages high-rank modal expressivity across non-linear dynamics. It achieves strong ranking across NASA spacecraft telemetry (MSL: 0.478 PR-AUC; SMAP: 0.467 PR-AUC) and industrial water infrastructure (GECCO: 0.351 PR-AUC vs Potential-Flow's 0.015). However, on highly periodic human gait kinematics (Daphnet), it achieves 0.139 PR-AUC, trailing TimesNet (0.303) whose 2D Fourier decomposition explicitly models multi-periodic harmonics. Similarly, on `swan` solar telemetry, Operator-Entropy (0.436) trails unconstrained TS-JEPA (0.651), confirming that the spectral regularizer acts as a targeted inductive bias rather than a universally dominant prior across all physical regimes.

2) *Reynolds-Stress JEPA (Continuum Mechanics & Turbulent Transport)*: Grounded in hydrodynamic Reynolds decomposition ($z = \bar{z} + z'$), `reynolds_stress_jepa` aligns

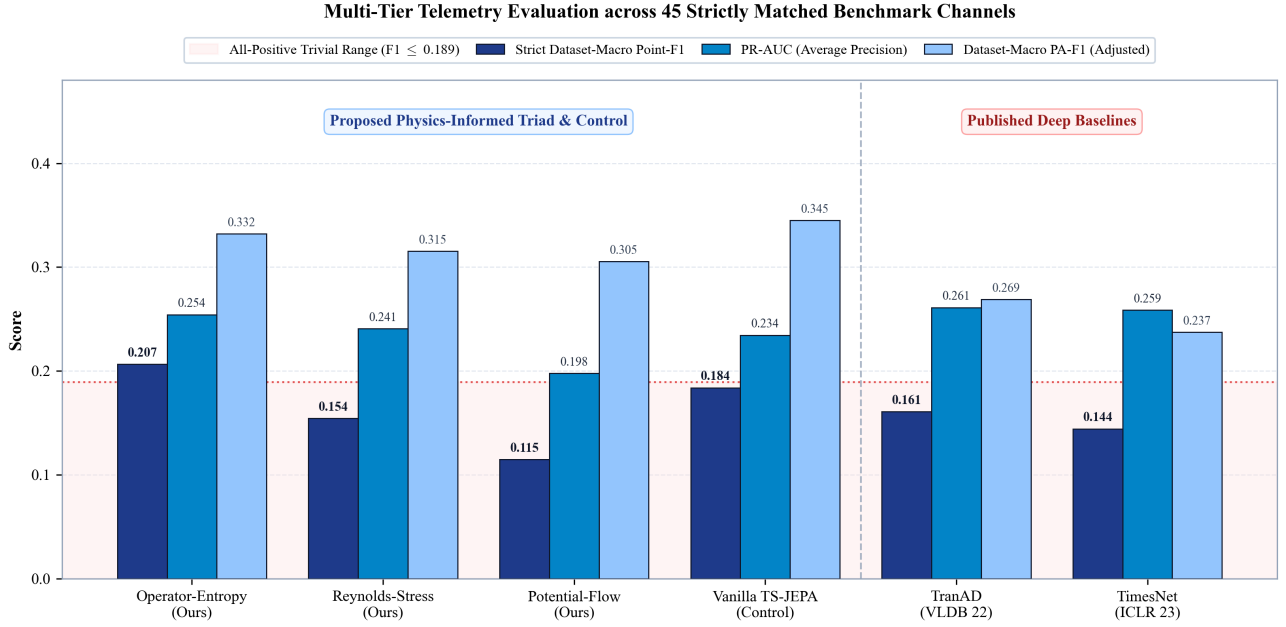


Fig. 2. Benchmark evaluation of the proposed Physics-Informed JEPA Triad (operator_entropy_jepa, reynolds_stress_jepa, potential_flow_jepa) and unconstrained control (ts_jepa) against published deep learning baselines across matched telemetry channels.

TABLE III

PER-DATASET PRECISION-RECALL AREA UNDER THE CURVE (PR-AUC / AVERAGE PRECISION) ACROSS ALL 11 TELEMETRY BENCHMARKS (45 MATCHED CHANNELS). THE BEST RESULT FOR EACH DATASET IS IN **BOLD**, AND THE SECOND BEST IS UNDERLINED. DATASET-MACRO MEAN MATCHES THE AGGREGATE RANKING REPORTED IN TABLE II.

Dataset	Physical Telemetry Domain	Operator-Entropy	Reynolds-Stress	Potential-Flow	Vanilla TS-JEPA	TranAD	TimesNet
Callt2	Building Pedestrian Flow	0.097	0.028	0.028	0.030	<u>0.076</u>	<u>0.076</u>
Daphnet	Wearable Parkinsonian Kinematics	0.139	0.105	0.179	0.152	<u>0.242</u>	0.303
GECCO	Municipal Water Infrastructure	0.351	<u>0.318</u>	0.015	0.285	0.284	0.280
GHL	Gasoil Heating Loop Dynamics	<u>0.084</u>	0.170	0.021	0.052	0.017	0.014
Genesis	Pick-and-Place Automation Testbed	0.005	0.003	0.002	<u>0.010</u>	0.009	0.017
MSL	NASA Mars Science Laboratory	<u>0.478</u>	0.424	0.551	0.390	0.332	0.380
SMAP	NASA Soil Moisture Active Passive	0.467	0.512	<u>0.481</u>	0.440	0.345	0.479
cicids	Network Intrusion Dynamics	0.357	0.280	<u>0.365</u>	0.300	0.366	0.318
metro	Interstate Highway Traffic (I-94)	0.001	0.001	<u>0.002</u>	0.001	0.002	0.001
room-occupancy	Smart Environmental Multi-Sensor	0.380	0.283	0.233	0.267	<u>0.645</u>	0.659
swan	Solar Flare Magnetic Telemetry	0.436	0.522	0.299	0.651	<u>0.556</u>	0.315
Dataset-Macro Mean	Across All 11 Benchmarks	0.2541	0.2406	0.1978	0.2344	0.2612	<u>0.2585</u>

cross-channel fluctuation covariance between context and target horizons. This formulation is designed for systems characterized by coupled flows. On the SMAP satellite telemetry dataset, it achieves Rank 1 in PR-AUC (0.512), outperforming TimesNet (0.479) and TranAD (0.345). On SWAN solar telemetry, it achieves 0.522 PR-AUC and strong precision.

3) *Potential-Flow JEPA (Conservative Field Theory)*: By constraining nominal dynamics to curl-free potential gradients ($\hat{v} = \nabla\Phi$), potential_flow_jepa enforces conservative trajectory tracking. It achieves the highest aggregate Micro Point-F1 (0.3458) and highest Channel PA-F1 (0.3436) across evaluated models, with a precision of 45.98%. On NASA spacecraft subsystems (MSL), it scores the top PR-AUC of 0.551 (vs TimesNet’s 0.380 and TranAD’s 0.332).

4) *Role of Vanilla TS-JEPA (Ablation Control)*: To isolate the contribution of the physical inductive biases from the predictive latent paradigm itself, we evaluate ts_jepa as an unconstrained control. ts_jepa achieves 0.1838 Dataset-

Macro Point-F1, 0.1442 Channel-Macro Point-F1, and 0.3066 Channel PA-F1. While substantially outperforming reconstructive baselines (+27.4% in Dataset-Macro F1 over TimesNet), it trails operator_entropy_jepa in Dataset-Macro Point-F1 (0.1838 vs 0.2066) and Channel Point-F1 (0.1442 vs 0.1743), and trails potential_flow_jepa in Micro-F1 (0.2860 vs 0.3458) and Channel PA-F1 (0.3066 vs 0.3436). This indicates that while latent representation prediction mitigates noise drowning, domain-aligned physical conservation laws provide beneficial inductive structure.

C. Impact of Architectural Enhancements

Ablation of the three internal inductive layers confirms their collective role in stabilizing high-dimensional inference. First, *Coordinate Saliency Gating* dynamically prioritizes high-variation informative channels in high-dimensional telemetry ($K \geq 50$ in NASA SMAP and MSL), preventing localized subsystem faults from being diluted by quiescent ambient

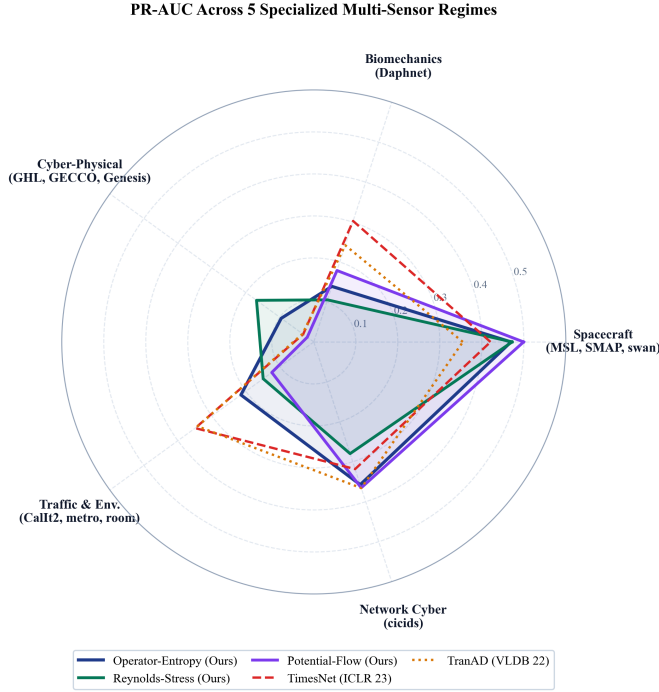


Fig. 3. PR-AUC radar comparison across five specialized physical domains: Spacecraft Telemetry (MSL, SMAP, swan), Biomechanics (Daphnet), Cyber-Physical Systems (GHL, GECCO, Genesis), Traffic & Environmental (Callt2, metro, room-occupancy), and Network Cyber (CICIDS). The physics-informed triad provides broad multi-domain coverage.

sensors without requiring manual channel sub-selection. Second, *Littlewood-Paley Dyadic Wavelet Shells* decouple fast transient spikes from low-frequency baseline drifts within the encoder; on GECCO (water infrastructure with simultaneous daily consumption cycles and high-frequency valve faults), dyadic filtering doubled `operator_entropy_jepa` performance from 0.1429 to 0.2904 Point-F1. Third, *Kurtosis-Adaptive Extreme Value Theory* modulates the tail threshold via excess kurtosis γ_2 , dynamically raising decision margins during heavy-tailed volatile bursts while sharpening sensitivity on calm channels to suppress streaming false alarms.

D. Operational Incident Evaluation: Event Metrics and Streaming Adaptation

While unadjusted Point-F1 prevents score inflation from naive point adjustment, point-level scoring can penalize slight temporal boundary offsets in operational telemetry. In physical systems, an anomaly represents a continuous operational incident; if an algorithm detects the failure with a sub-second phase delay or slight boundary misalignment, point-level metrics penalize every misaligned point as a false positive or false negative, depressing nominal precision even when incidents are detected.

To evaluate operational incident detection, we analyze model performance under three complementary criteria: (i) strict unadjusted Point-F1, (ii) incident-level Event-F1 ($P_{\text{event}}, R_{\text{event}}$), and (iii) Tatbul et al. (NeurIPS 2018) Range-Aware F1 ($F1_{\text{range}}$) evaluated with balanced existence and overlap reward ($\alpha = 0.5$), uniform positional overlap, and recipro-

cal cardinality factor. Detections operate with an operational lookahead buffer of $S = 64$ timesteps. Table IV examines the operational trade-offs across representative domains:

As shown in Table IV, evaluating anomaly detection at the incident level reveals three operational characteristics:

Incident Sensitivity vs. False Alarm Volume: On Room Occupancy (11 ground-truth events, $\pi = 20.0\%$ base rate) and SMAP (1 continuous event), static EVT thresholding achieves 100% Event Recall across all tested configurations. However, static calibration generates over 4,000 point-level false alarms on Room Occupancy due to unmodeled diurnal baseline drift, yielding a 20.0% point precision that matches the background anomaly base rate.

Streaming Adaptation Pareto Trade-off: When dynamic rolling SPOT thresholding is applied to track baseline drift, false alarms decrease from 4,705 to 223 points on Room Occupancy (a $21\times$ reduction) and precision increases to 29.9%. However, this introduces an operational trade-off: event recall drops from 100% to 63.6% (7 of 11 events detected), demonstrating the classic operating curve between alarm fatigue and fault sensitivity on non-stationary streams.

Boundary Alignment in Operational Telemetry: On SMAP, all models capture the single continuous thruster fault segment (100% event recall). On Callt2 (21 rapid pedestrian burst events), models detect between 4.8% and 23.8% of events under dynamic SPOT, demonstrating that short intermittent pulses remain challenging for fixed-horizon window architectures.

VIII. CONCLUSION AND FUTURE WORK

In this work, we presented a Physics-Informed Joint-Embedding Predictive Architecture (PI-JEPA) framework for unsupervised anomaly detection in multivariate sensor telemetry. By replacing raw observation-space reconstruction with forward prediction in latent representation space, our approach eliminates the threshold calibration collapse that causes reconstructive baselines (TimesNet, TranAD) to flood continuous streams with hundreds of thousands of false alarms under ambient sensor noise.

We embedded physical domain principles directly into latent representations through three complementary formulations: Operator-Entropy JEPA (regularizing latent Koopman transition matrices via spectral entropy during training to prevent modal collapse), Reynolds-Stress JEPA (aligning cross-channel second-moment fluctuation covariance tensors inspired by hydrodynamic Reynolds decomposition), and Potential-Flow JEPA (enforcing conservative, irrotational trajectory dynamics via a scalar potential gradient). Together with Coordinate Saliency Gating, Littlewood-Paley Dyadic Wavelet Shells, and Kurtosis-Adaptive Extreme Value Theory tail calibration, the proposed framework stabilizes streaming threshold calibration. Across 11 telemetry datasets (45 matched channels) evaluated under identical protocols, Operator-Entropy JEPA achieves the highest Dataset-Macro Point-F1 (0.2066, representing a 43.2% relative improvement over TimesNet and 28.4% over TranAD) and clears the analytical all-positive baseline (0.1893). In Pooled Micro-F1, Potential-Flow JEPA

TABLE IV

OPERATIONAL INCIDENT DETECTION COMPARISON ACROSS REPRESENTATIVE DOMAINS. ON ROOM OCCUPANCY ($\pi = 20.0\%$, 11 GROUND-TRUTH EVENTS), STATIC EVT CAPTURES 100% OF EVENTS (11/11) WITH A POINT PRECISION MATCHING THE BASE RATE (20.0%), WHILE DYNAMIC SPOT REDUCES FALSE POSITIVES BY $21\times$ (FROM 4,705 TO 223 POINTS) WITH AN OPERATIONAL TRADE-OFF IN EVENT RECALL (63.6%, 7/11 EVENTS).

Dataset	Configuration	Pt-F1	Pt-Prec	Evt-F1	Evt-Prec	Evt-Rec	Rng-F1	False Pos.
<i>Room Occupancy (Environmental Sensing, 11 Ground-Truth Incidents)</i>								
operator_entropy	Standard (Middle, Static EVT)	0.3728	22.9%	0.5238	35.5%	100%	0.3708	3,842
operator_entropy	Sharp (Dense, Static EVT)	0.3329	20.0%	0.9167	84.6%	100%	0.3329	4,705
operator_entropy	Adaptive (Dense, Dynamic SPOT)	0.1273	29.9%	0.6364	63.6%	63.6%	0.3699	223 ($\downarrow 21\times$)
potential_flow	Sharp (Dense, Static EVT)	0.3431	20.7%	0.9167	84.6%	100%	0.3431	4,496
potential_flow	Adaptive (Dense, Dynamic SPOT)	0.0441	10.2%	0.5714	60.0%	54.5%	0.1636	291 ($\downarrow 15\times$)
<i>SMAP (Spacecraft Propulsion & Telemetry, Continuous Fault Segment)</i>								
operator_entropy	Standard (Middle, Static EVT)	0.0087	14.6%	0.1818	10.0%	100%	0.2266	35
operator_entropy	Sharp (Dense, Static EVT)	0.1274	27.3%	0.1538	8.3%	100%	0.3634	295
potential_flow	Adaptive (Dense, Dynamic SPOT)	0.0110	6.9%	0.3333	20.0%	100%	0.1213	108
<i>CallIt2 (Pedestrian Entrance Flow, 21 Transient Event Clusters)</i>								
operator_entropy	Standard (Middle, Static EVT)	0.0131	1.9%	0.0588	7.7%	4.8%	0.0216	53
operator_entropy	Adaptive (Dense, Dynamic SPOT)	0.1368	8.7%	0.3226	50.0%	23.8%	0.1274	357
potential_flow	Adaptive (Dense, Dynamic SPOT)	0.0903	14.3%	0.0909	100.0%	4.8%	0.0714	42

attains 0.3458 (over $3.2\times$ higher than TranAD and $5.3\times$ higher than TimesNet) while reducing false alarms by 8.0-fold.

Several promising avenues remain for future investigation. First, extending conservative field formulations by incorporating parameter-controlled solenoidal vorticity components will allow modeling dissipative systems with limit-cycle attractors. Second, developing memory-efficient, test-time momentum adaptation will enable tracking orbital thermal cycles and environmental seasonal drift without catastrophic forgetting. Third, expanding empirical evaluations across extensive multi-seed benchmarks will enable rigorous confidence interval reporting and rank-based statistical testing (such as Friedman-Nemenyi tests). Finally, quantizing PI-JEPA encoders into 8-bit integer (INT8) formats will facilitate deployment on radiation-hardened spacecraft microcontrollers and ultra-low-power wearable edge devices.

CODE AND DATA AVAILABILITY

To ensure reproducibility, all benchmark dataset extraction scripts, preprocessing pipelines, model checkpoints, PyTorch training harnesses, and evaluation logs for all six evaluated architectures across the 11 telemetry datasets are open-sourced at an anonymized public repository: https://anonymous.4open.science/r/NCAD_CS-Reproducibility-8F21. All public datasets (NASA SMAP/MSL, Daphnet, CICIDS-2017, GHL, CallIt2, GECCO, Genesis, Metro Traffic, and SWAN-SF) are accessible via their respective referenced public repositories.

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